Visualizing Twitter Opinions with Social Network Analysis: Vaccine Debates amid the COVID-19 Beta and Omicron pandemics

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Abstract

COVID-19 vaccination rate remained globally low despite governments' ongoing efforts to encourage vaccination. Sentiment analysis and social network analysis were employed on tweets collected on selected dates during the Beta and Omicron pandemics. It is found that the data exhibit real-world properties. People became more polarized and negative towards vaccination during the Omicron pandemic. Unaccredited authors exerted immense influence on anti-vaccination clusters, which became more concentrated after a year. To increase vaccination rate, it is suggested government officials enhance their presence in these platforms or social media companies reconstruct the information diffusion mechanism to cultivate the interaction anti-vaccination clusters with other communities.

1. Introduction

COVID-19 became a global pandemic since its first emergence in December 2019. Among all public health measures like social distancing and mandatory mask-wearing in public, vaccination was deemed the most effective by most governments (Priniski and Holyoak 2022). Crippled by the mutations of the coronavirus, many countries struggled with containing the pandemic Until March 2022, nearly 500 million cases were confirmed, with 1.4 million daily new cases and 6 million deaths (Team 2022). Lamentably, only 59% of the total population were fully vaccinated, below the predicted threshold of 60–70% (Aschwanden 2021). In fact, skepticism of COVID-19 vaccines undermined the public's inclination to get vaccinated. One of the commonest channels people propagate their disbelief of COVID-19 vaccines is social media like Twitter. Not only do like-minded users easily form coalitions reinforcing their beliefs, but circulating misinformation also overwhelms the correct information, potentially misleading people from abiding by public health measures (Benoit and Mauldin 2021).

This study analyzes the tweeting networks during the Beta and Omicron pandemics (the networks are referred to as "Beta" and "Omicron" for simplicity). Employing social network and sentiment analyses, it aims to test two hypotheses: (i) Beta and Omicron exhibit networking differences; (ii) Both mimic real-world networks. Last, it discusses the challenges Twitter poses to facilitate vaccination.

2. Methods

Data Collection

TWINT was used to scrape tweets mentioning "vaccine" or "vac" (including hashtags) during the Beta and Omicron pandemics. As both waves took place in January, "7" was randomly generated to scrape tweets on January 7, 2021 (Beta) and 2022 (Omicron). Only tweets in English were collected. After extracting the tweets mentioning, retweeting, or replying to others, directed edge lists were produced. While the study primarily analyzes the scalar structure of the network, the direction of the connections offers secondary insight to the influence of individual authors.

Data Analysis

Besides visualization on Gephi, the networks were analyzed with R. First, different measurements of the networks were calculated and tested for their differences with Kruskal-Wallis tests. Positive results would lead to the acceptance of the first hypothesis. Second, the small-world index and logarithmic degree distribution were evaluated to identify any real-world properties in them. The second hypothesis would be accepted if the index was greater than 1 and the graph formed a straight line. Third, sentiment analysis was employed on the tweets with AFINN and NRC dictionaries (the logarithmic values of NRC scores are used for better visualization of the dispersion of emotions).

Supervision

All methods were carried out in accordance with relevant guidelines and regulations of the Department of Geography, Faculty of Social Sciences, University of Hong Kong. All experimental protocols were approved by the Department of Geography, Faculty of Social Sciences, University of Hong Kong. Since the study collects publicly available data on the Internet, collection of informed consents from all subjects is inapplicable. Only information published publicly by the subjects would be analyzed.

3. Results

Structure

Table 1 summarizes the dataset. 39882 and 74506 tweets were collected for the Beta and Omicron pandemics. Unique authors are users who ever tweeted in the examined period. Among the authors of these tweets (33085 and 30427), only a half of them interact with other users (18433 and 19618) whereas the remaining authors are isolated. Sources and targets contribute to the nodes in the networks. Besides, the sum of sources and targets was slightly higher than the number of nodes in the networks because some users are both sources and targets themselves (Fig. 1 visualizes the concept). On the other hand, with a similar number of authors and doubled collected tweets in Omicron, users tweeted more frequently in Omicron than in Beta. With higher directed network diameter and average path length, tweets resulted in longer chains, extended to 2 more tweets before being concluded.

Figure 2 visualizes the networks with ForceAtlas2 whereas Table 2 summarizes different measurements. Kruskal-Wallis test was done to test whether the networks followed the same distributions. With p-values all smaller than 0.01, the null hypothesis that no difference existed is rejected, and the alternative hypothesis that Omicron differed from Beta is accepted.

Degrees, in-degrees, and out-degrees increased in Omicron due possibly to a bigger network. Though eccentricities, closeness centralities, and harmonic closeness centrality dropped, Omicron recorded higher clustering and lower modularity. That said, more interactions occurred between subcommunities within the network. Notably, betweenness centralities and eigenvector centralities increased in Omicron; the presence of “brokers” and influential users might contribute to the aggregation of the network.

Real-world properties

Figure 3 visualizes the networks with CircularLayout. Arranged counterclockwise in descending order of degree, both networks manifest similar distribution of edges, where nodes with high degrees are hubs. The denser edges in Omicron may be attributed to the larger size of the network. The small-world index was then calculated (Humphries and Gurney 2008) (Table 3). Since the small-world model was proposed to describe binary graphs, average clustering coefficients and path lengths were recomputed assuming the networks to be undirected (Watts and Strogatz 1999). It is found that the average path lengths of both networks are smaller than that of their equivalent random network whereas the clustering higher. Both networks exhibit small-world properties. Since they follow straight lines in the logarithmic graphs of the degree distributions (Fig. 4), they also exhibit scale-free properties.

Sentiment Analysis

Words relevant to vaccination were the commonest words in the tweets, e.g., “get”; while vaccine brands like “pfizer” and “moderna” were frequently mentioned in Beta, “booster” seemingly replaced the discussion of which vaccines to get (Fig. 5). From the Beta bigram cloud, popular words were either related to the government, e.g., “uk parliament” and “government sign,” or skepticism of vaccination, e.g., “long term,” and “side effect.” In Omicron, skepticism of vaccination persisted, predominated by information about vaccination like “slots found” or “centers jan.” (Fig. 6)

In subsequent sentiment analyses, tweets manifested profound antagonistic sentiments in both networks. Despite a lower range, Omicron saw decreased mean AFINN scores and increased standard deviation; that said, the sentiments became more polarized, shifting slightly to negativity (Fig. 7; Table 4). Emotionally, anticipation and trust plunged by 26.06% and 6.37% while fear and sadness escalated by 24.13% and 8.02% (Fig. 8; Table 5).

Popular targets were celebrities while popular sources were ordinary netizens (Table 6). UK and US government accounts had the highest indegrees in Beta; they were replaced by news media in Omicron. In both networks, popular sources usually included anticipation and trust towards vaccine; per examination, all 6 users were pro-vaccine. Influential users with the highest eigenvector centralities were always those with the highest in-degrees. On the other hand, two betweenness centralities were computed. With the highest directed betweenness centralities, users’ tweets could disseminate information across subcommunities. For example, @bbcquestiontime was highly penetrative across Beta, whereas @tenebra99 attacked pro-vaccine users, receiving diverse feedback. With the highest undirected betweenness centralities, however, users were bridges of information. If the direction of information flow is neglected, users like Boris Johnson and Emmanuel Macron were the accounts interacting with important information; though they did not tweet during the observed period, the tweets they were mentioned carried considerable significance to the networks.

4. Discussion

In Omicron, users became more active in expressing their opinions towards vaccines on Twitter due possibly to the aggravated pandemic. While some users became more aware of the importance of mass vaccination, some developed feelings toward the ongoing and heightened public health measures. Users might get comfortable with interacting with other users within their subcommunities, which explained the doubled amount of collected tweets and lengthened network diameter. On the other hand, users tended to become more influential in Omicron with increased eigenvector centralities and betweenness centralities. Since tweeting networks demonstrate small-world and scale-free properties, not only are “hubs” the sources of information to distant users who would retweet to their own communities, but new users also preferentially attach to these “hubs.” Given that 72% of adult internet users searched online for health issues (Fox 2009), although the sample concerns only the tweeting behaviors of English-speaking Twitter users, the influence of Twitter should not be overlooked as it extends from the virtual space to reality.

Relevant to shrinking subcommunities is the aggravated polarization of sentiments. With aggregating subcommunities, opinions on Twitter became less diverse due possibly to the influence of hubs. Nevertheless, negative sentiments expressed in Omicron might differ from in Beta. In Beta, Twitter was flooded with conspiracy theories regarding the side effects of vaccines; while such worries persisted among anti-vaccine users in Omicron, as implied by decreased anticipation and increased fear, fundamental to the negative sentiments might be anti-pandemic fatigue toward an unresolved pandemic (World Health Organization 2020). Low vaccination rate aside, mental health issues in people are also perplexing.

Another interesting finding is that betweenness centralities gave different results as the networks were considered “directed” and “undirected.” Government officials and celebrities were important “brokers” when the graph was “undirected” with many in-degrees; however, if considering the direction, these accounts were beaten by ordinary netizens who replied and mentioned more people (out) and thus receive more replies (in). An example of @tenebra99 also shows that some users establish their influence by mentioning users from other subcommunities. Since the calculation of centralities for undirected and directed graphs differ, the actual mechanism of how netizens derived their influence comes to question; however, this poses an incentive for celebrities and government officials to make good use of their “degrees” in social media to facilitate public health information.

In conclusion, this study finds that Twitter network structure cripples the facilitation of vaccination, underestimated by most scholars and governments. Discussions on COVID-19 vaccinations are predominated by influential yet unaccredited authors responsible for forming polarized subcommunities. Persistently low vaccination rates may be ascribed to the misinformation on vaccines which can be rapidly transmitted to distant users; these subcommunities are nevertheless aggregated with more extreme sentiments further polarizing the Internet. Besides, netizens became more negative towards COVID-19; thus, not only were the anti-vaccination sentiments intensified, but the uncontrolled Omicron pandemic resulted from low vaccination rates also indirectly discouraged the public. It is therefore advised that the government should make better use of online communities to enhance the penetration of pro-vaccination information. Instead of monitoring negative tweets, Twitter should aim to revise the algorithm so information on vaccination can be more frequently introduced to aggregated subcommunities otherwise dominated by a few influential users.

Declarations

Data availability

All data generated or analyzed during this study are included in this published article and its supplementary information files.

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Tables

Table 1: Summary of Dataset.

	Beta	Omicron
Collected Tweets	39882	74506
Number of unique authors	33085	30427
Number of unique sources in edges	18433	19618
Number of unique targets in edges	19118	25001
Nodes	36821	43499
Edges	36321	45501
Directed network Diameter (undirected)	3 (29)	5 (33)
Directed average path length (undirected)	1.061 (8.854)	1.142 (10.526)
Directed average clustering coefficient (undirected)	0.013 (0.072)	0.021 (0.106)

Table 2: Summary of measurements of Networks.

Parameter	Beta:	Omicron:	Kruskal–Wallis Statistics	p-value
	Mean (Standard Deviation)	Mean (Standard Deviation)		
Degree	1.973 (5.723)	2.092 (4.086)	233.5	1.04E-52
Indegree	0.986 (5.543)	1.046 (3.542)	296.6	1.81E-66
Outdegree	0.986 (1.939)	1.046 (2.413)	39.9	2.70E-10
Eccentricity	0.528 (0.554)	0.483 (0.567)	170.7	5.28E-39
Closeness centrality	0.492 (0.494)	0.442 (0.491)	212.4	4.15E-48
Harmonic closeness centrality	0.494 (0.495)	0.444 (0.492)	212.3	4.26E-48
Betweenness Centrality	0.063 (3.492)	0.169 (9.642)	28.1	1.17E-07
Clustering	0.013 (0.076)	0.021 (0.097)	301.8	1.35E-67
Eigenvector centrality	0.002 (0.011)	0.004 (0.014)	4449.3	0
Modularity (resolution=5)	3922.318 (1897.81)	2143.129 (1952.806)	14381.8	0

Table 3: Small world index of Beta and Omicron. A network can be said "small-world" if its small-world index is higher than 1.

Target	Small-world Index	Average clustering coefficient (average clustering coefficient of an equivalent random network)	Average path length (average path length of an equivalent random network)
Beta	2264.389	0.072 (5.431e-05)	8.854 (15.123)
Omicron	3107.163	0.106 (4.598e-05)	10.526 (14.187)

Table 4: Summary Statistics of AFINN Sentiment Analysis.

	Minimum (Adjusted)	Maximum (Adjusted)	Median (Adjusted)	Interquartile range (Adjusted)	Mean (Adjusted)	Standard Deviation (Adjusted)
Beta	-72 (3.33)	21.5 (4.8)	0 (4.61)	2.5 (0.025)	0.081 (4.61)	1.92 (0.021)
Omicron	-30 (4.25)	32 (4.88)	-0.5 (4.6)	5 (0.05)	-0.416 (4.6)	3.48 (0.035)

Table 5: Summary Statistics of NRC Sentiment Analysis.

Color	Emotion	Beta	Omicron
	Anticipation	34.21%	8.15%
	Trust	29.97%	23.6%
	Fear	17.37%	41.5%
	Joy	9.83%	8.43%
	Surprise	4.02%	2.23%
	Anger	1.56%	4.83%
	Sadness	1.54%	9.56%
	Disgust	1.5%	1.7%

Figures

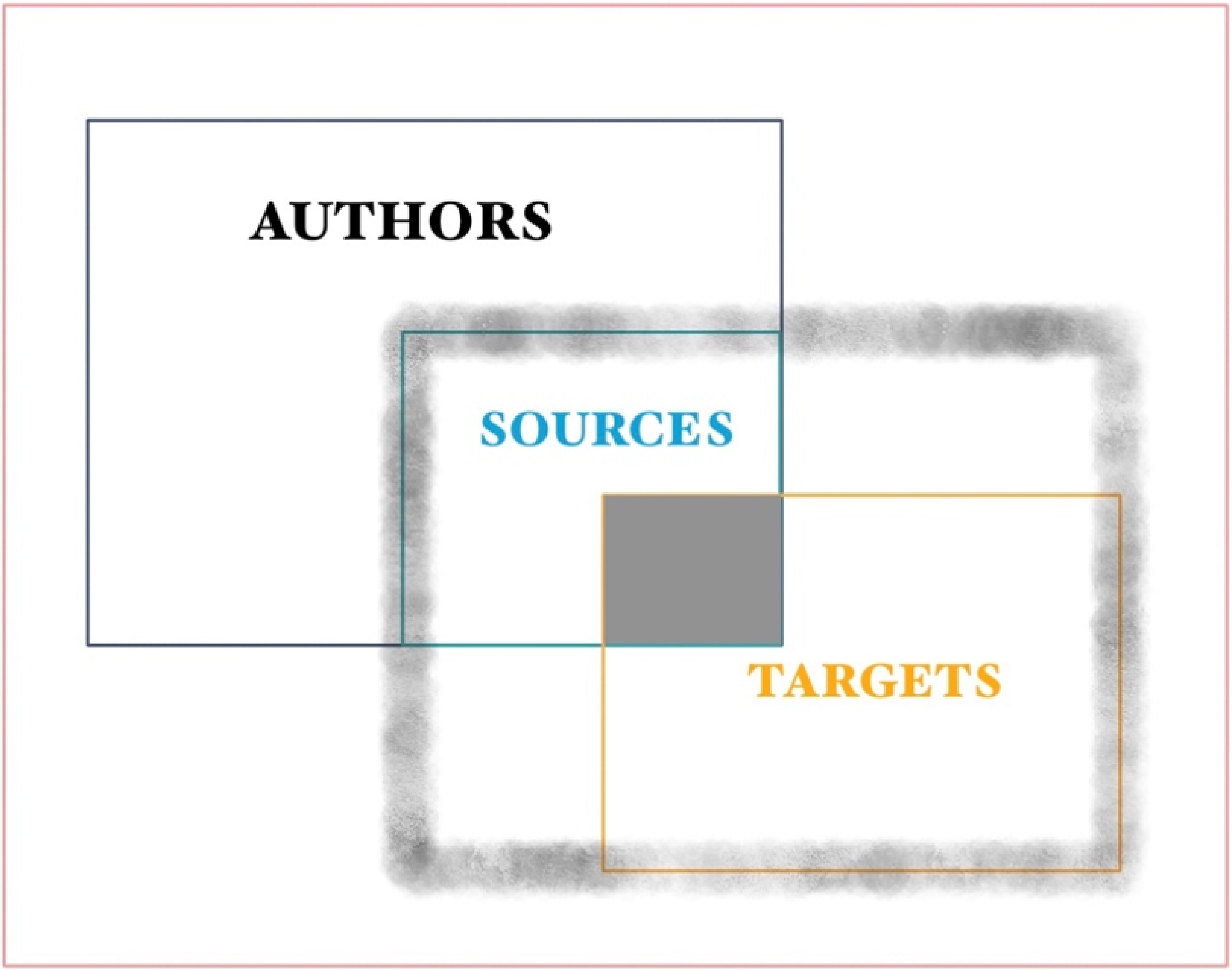

Figure 1

Visualizations of the distinctions between tweets, authors, sources, targets, nodes, and edges. Among all the unique "authors" of the collected tweets, "sources" form edges with other users. "Targets" are subjects of the interactions of the "sources;" while some of them are "authors," some of them do not contribute to the collected tweets. The shaded square represents users who are "sources" and "targets" themselves. Inside the murky gray box are the nodes in the networks, whereas the "authors" who have not interacted with other users are excluded.

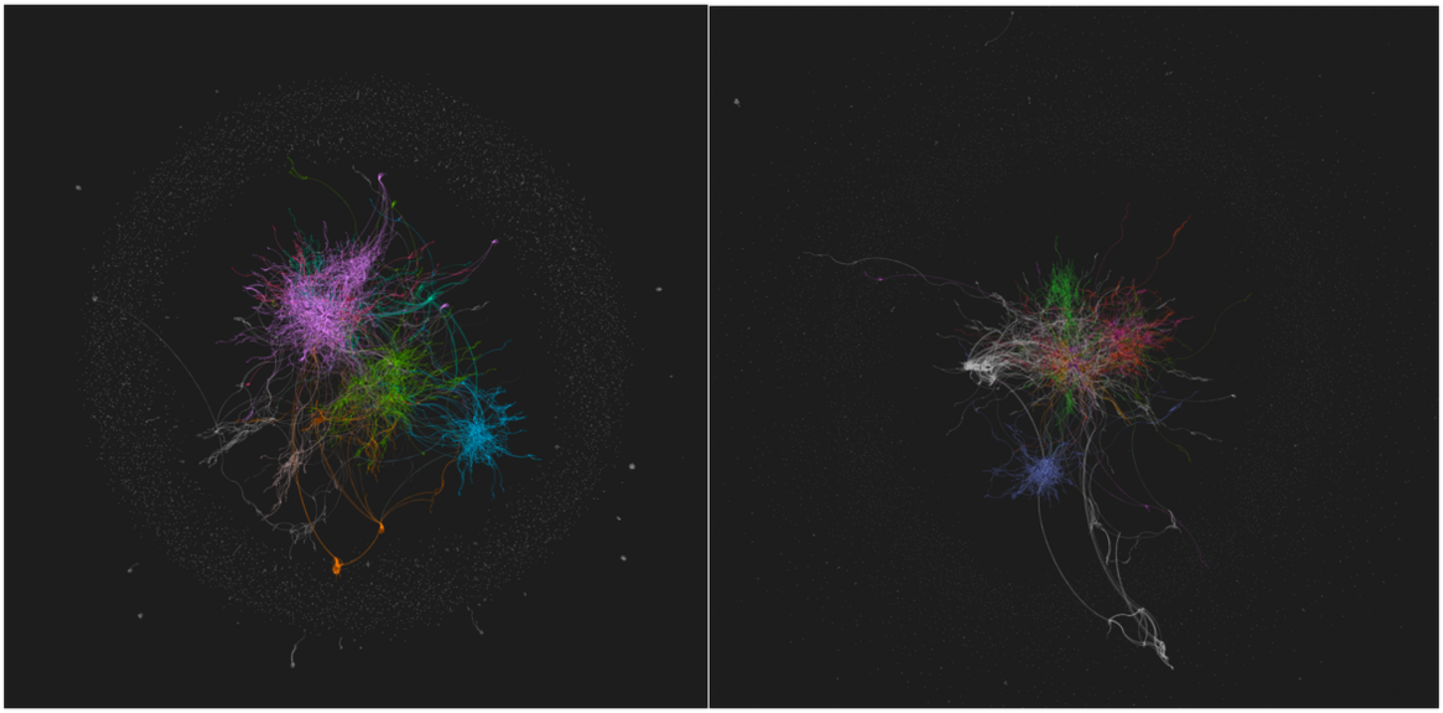

Figure 2

Graph networks of Twitter users during the Beta (left) and Omicron (right) pandemics, generated by ForceAtlas2. Sizes and colors of nodes correspond to their degree and modularity class. Hubs are dissuaded.

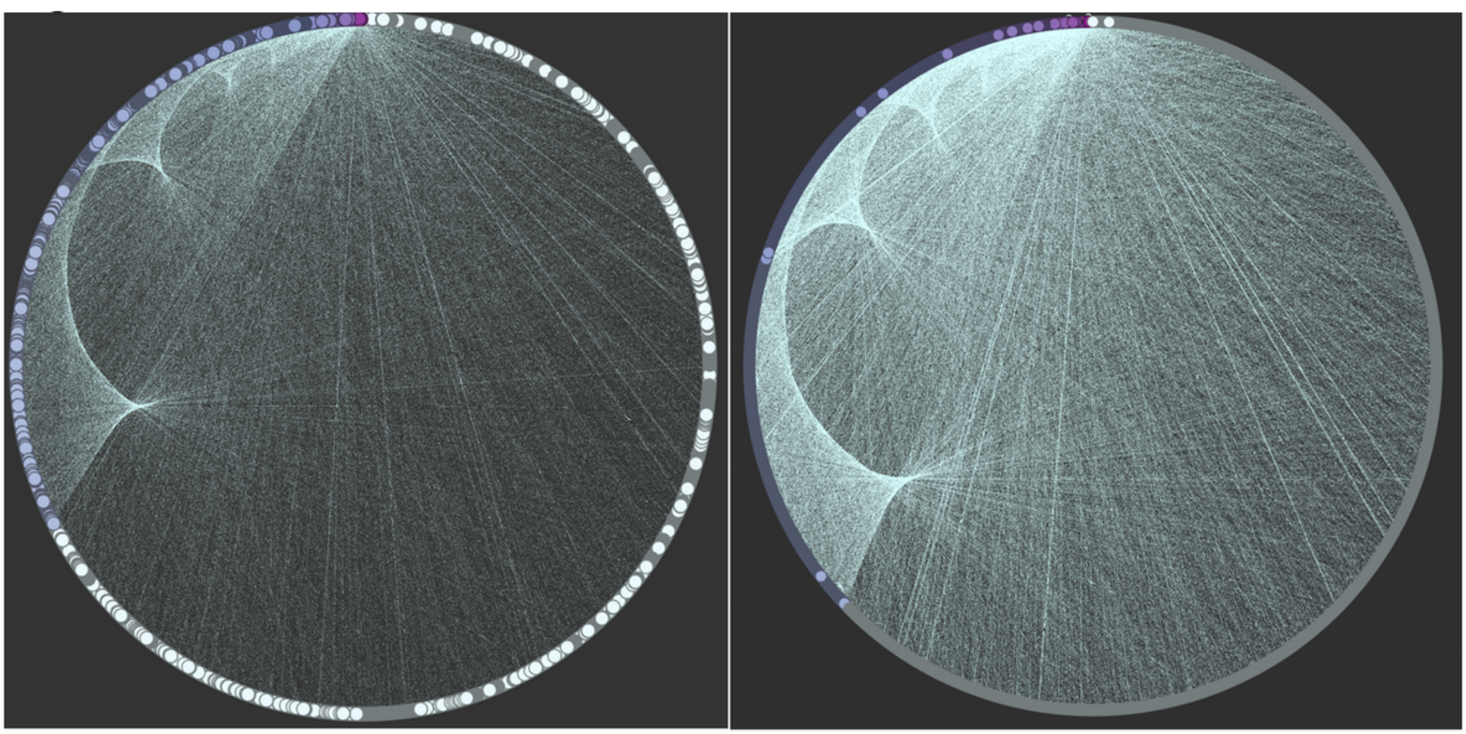

Figure 3

Graphnetworks of Twitter users during the Beta (left) and Omicron (right) pandemics, generated by circular layout. Color depth of nodes correlates with their degrees.

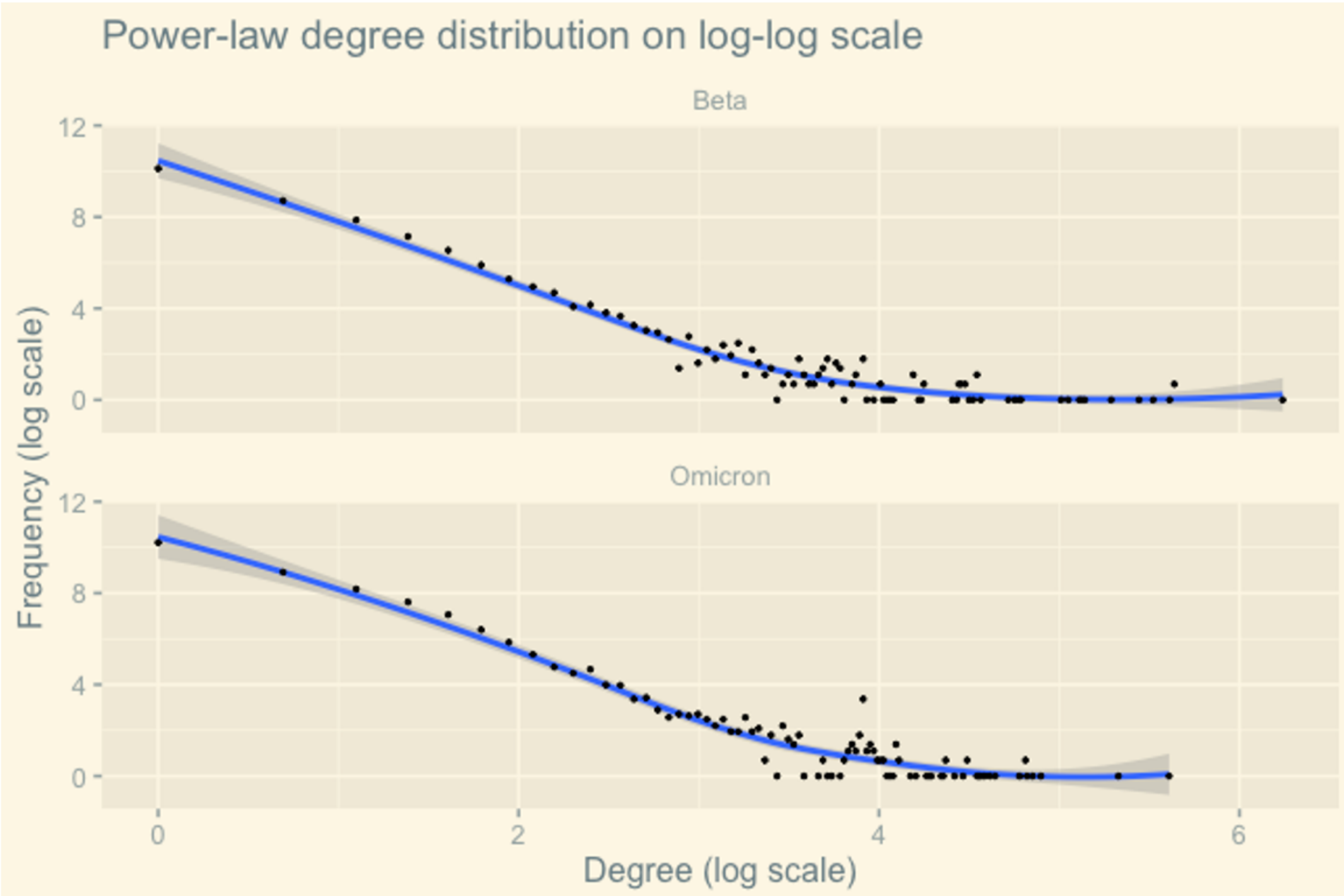

Figure 4

Power-law degree distribution on log-log scale of Beta and Omicron.

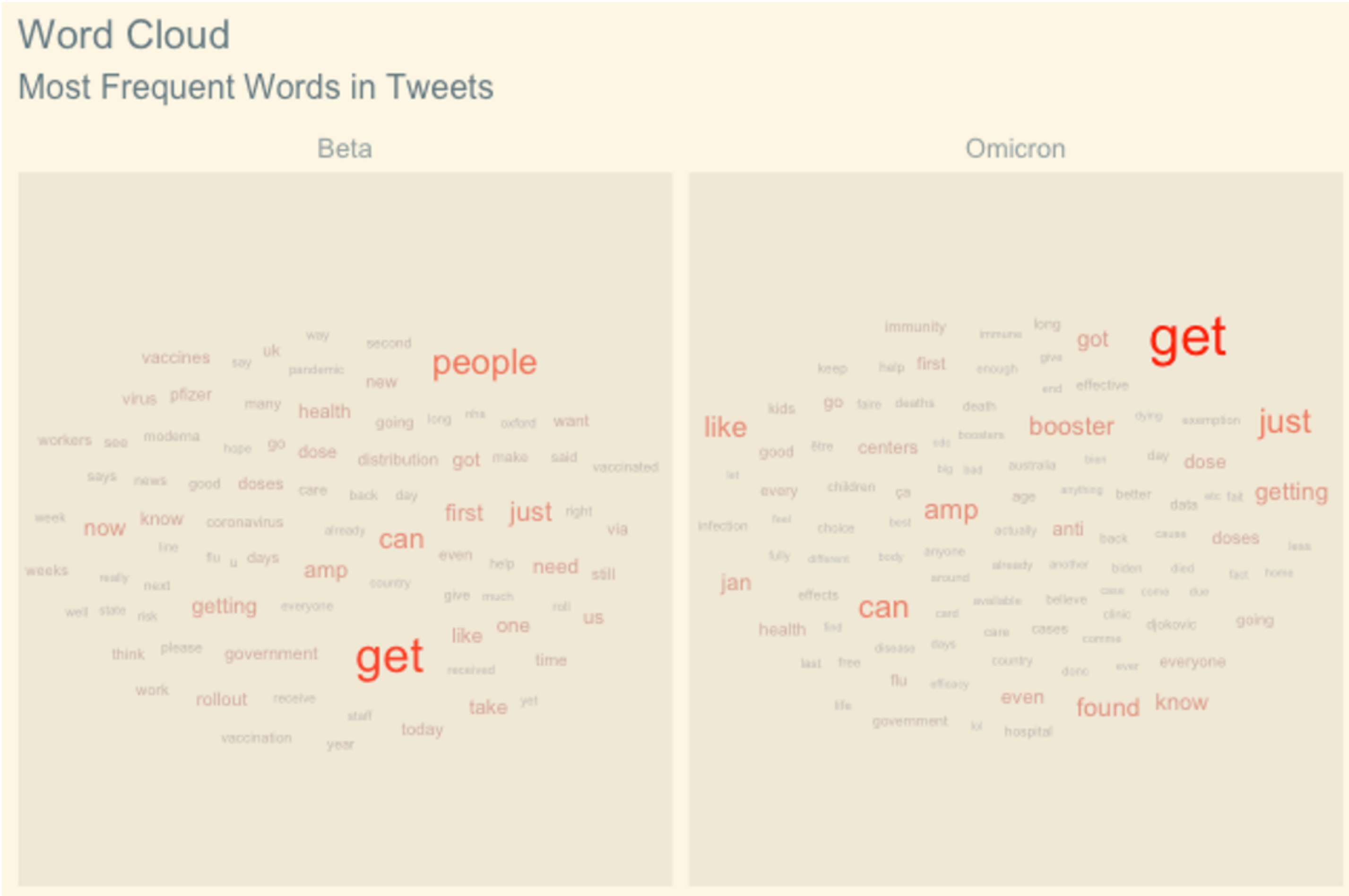

Figure 5

Word Cloud of Most Frequent Words in Beta and Omicron.

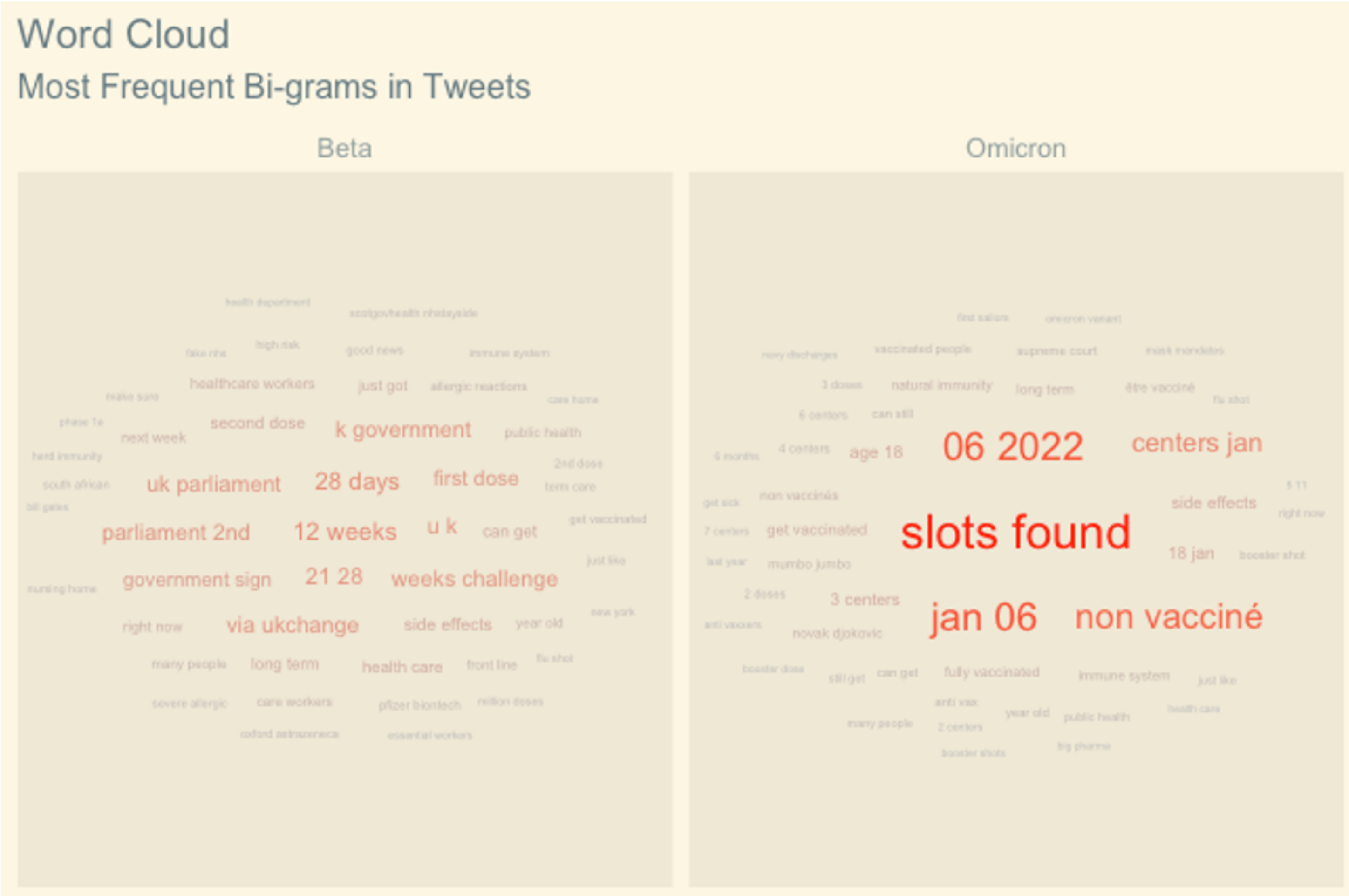

Figure 6

Word Cloud of Most Frequent Bigrams in Beta and Omicron.

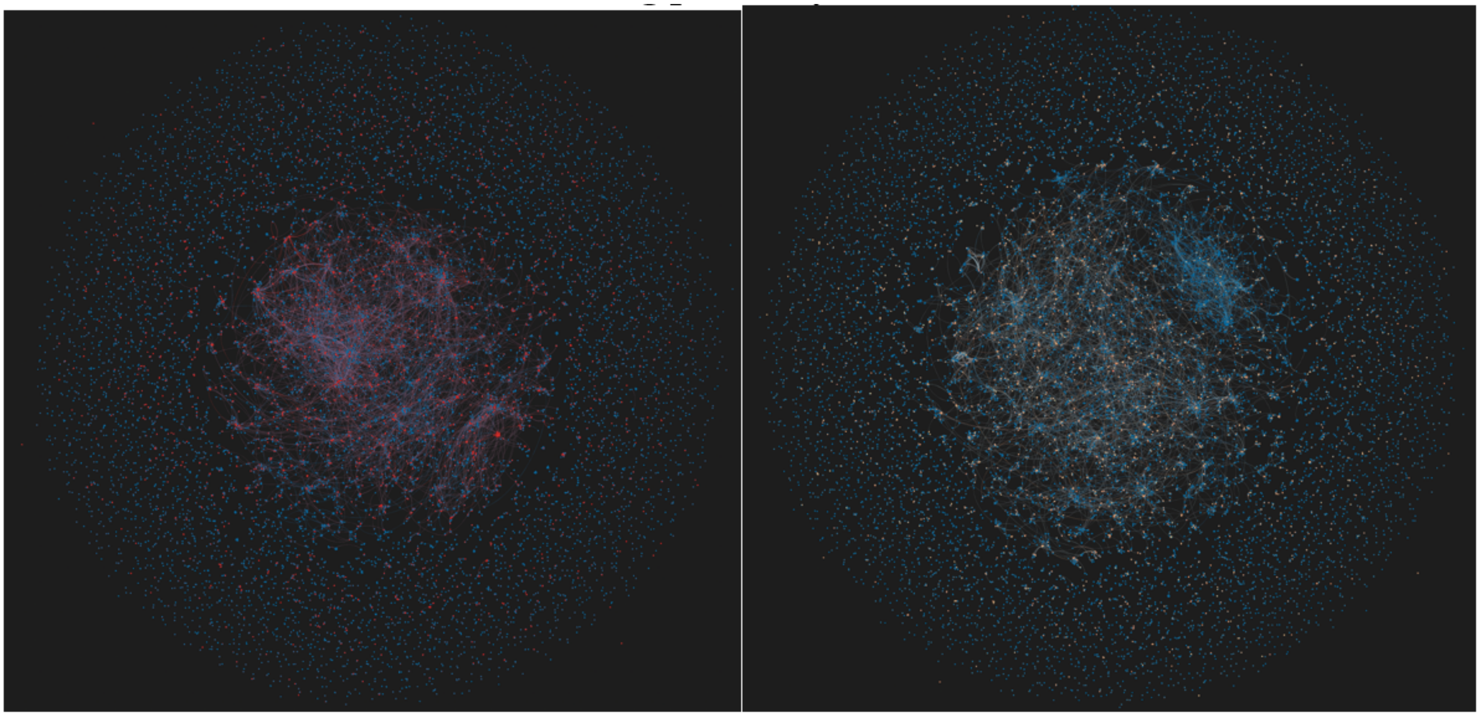

Figure 7

Graphnetworks of Twitter users during the Beta (left) and Omicron (right) pandemics, generated by OpenOrd. Sizes and colors of nodes correspond to their eigenvector centrality and logarithmic AFINN sentiment scores. Across the spectrum from blue to red correlates increasing positivity.

A picture containing star, dark, outdoor object, night sky Description automatically generated
A picture containing dark, night sky Description automatically generated

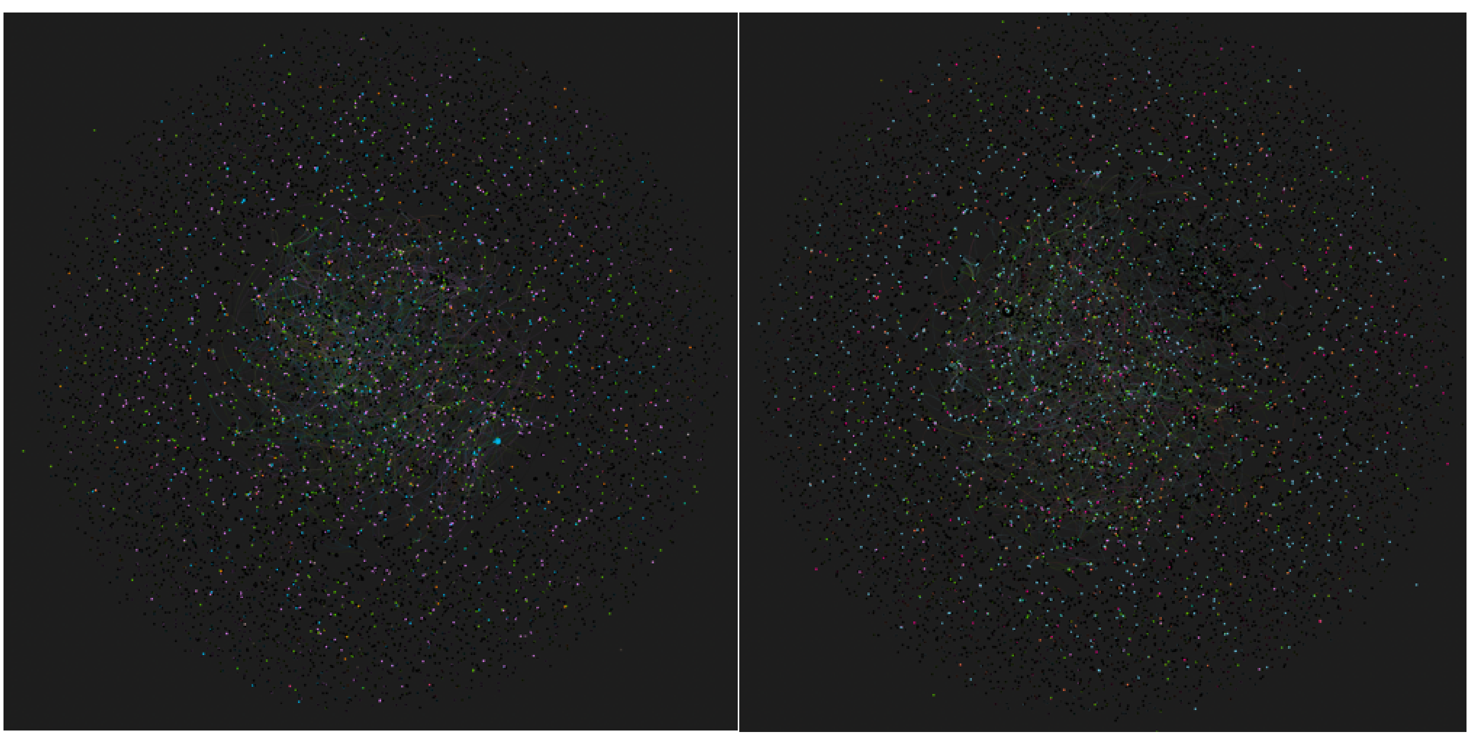

Figure 8

Graphnetworks of Twitter users during the Beta (left) and Omicron (right) pandemics, generated by OpenOrd. Sizes and colors of nodes correspond to their eigenvector centrality and the predominant sentiment in their tweets.

Supplementary Files

This is a list of supplementary files associated with this preprint. Click to download.

- betanew.csv
- omicronnew.csv